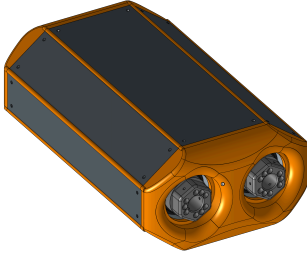
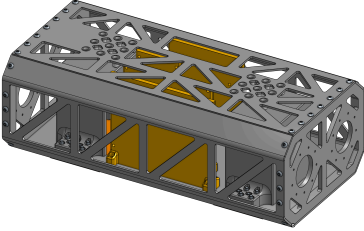
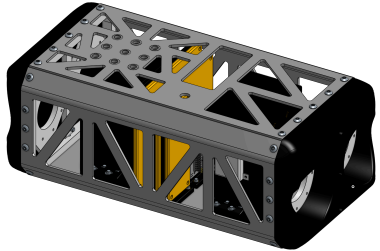


UMR Body Documentation

Body Versions:

V1	V2	V3
		
Pros: - Small - Nice Profile - Very light Cons: - Did not fit electronics - All 3D printed	Pros: - Fits the electronics - Room for air flow - Aluminum for strength Cons: - Too large - Heavy	Pros: - Small - Fits everything - Aluminum for strength - Lighter

Design Process:

The primary goals of the body were for it to have the following characteristics:

- Lightweight
- Ridgid
- Fits all electronics
 - 2 Batteries
 - 1 Jetson Nano
 - 3 PCBs of varying sizes
- Manufacturable on campus
- Must be able to mount “accessories”
- Jetson Nano ports must be accessible from the outside

Decisions made based on the listed goals include:

- Creating a mostly rectangular side profile with small chamfered corners to maximize internal volume while getting rigidity from bent aluminium side panels
- Create as much of the body as possible out of $\frac{1}{8}$ " aluminum plate for cost and ease of manufacturing
- Add lots of lightning to the aluminium to reduce weight
- Add an array of standardized mounting points using rivet-nuts
- Include milled aluminum end caps to hold other plates and add rigidity

Final Design:

Specifications:

Weight: 6.753 lbs

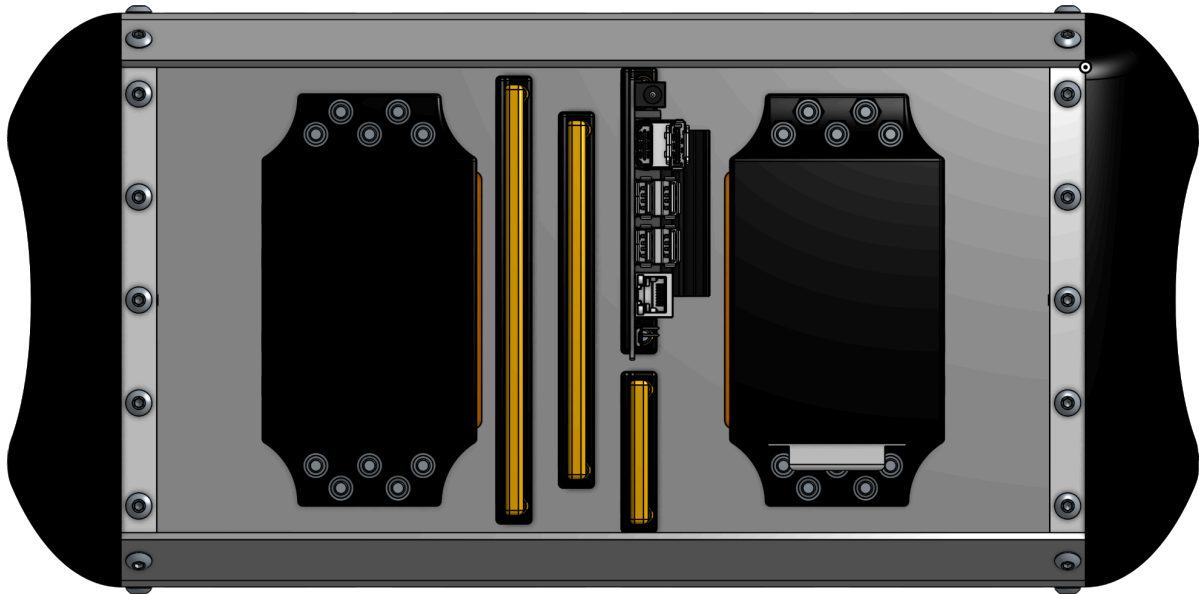
Dimensions:

L: 16.688 in

W: 8.202 in

H: 6.327 in

Layout of electronics



Full Assembly Drawing

